BlueBank – Secure Azure Financial Architecture
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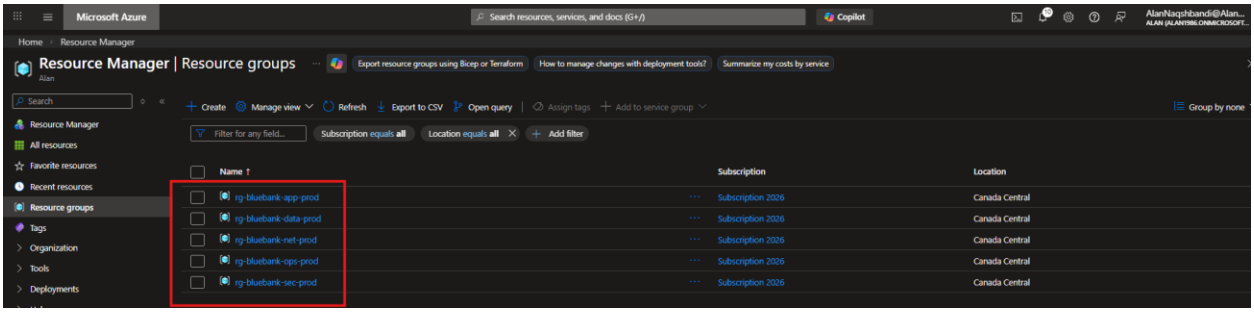
Role: Azure Security Engineer (Portfolio Project) 2026

Executive Summary

This document provides structured evidence of the secure deployment and validation of the BlueBank Azure production architecture. It demonstrates a financial-grade design with a public application layer and fully private backend services, implemented using network segmentation, Private Endpoints, Private DNS, group-based RBAC, Managed Identity, Microsoft Defender for Cloud protections, and centralized logging and alerting for operational readiness.

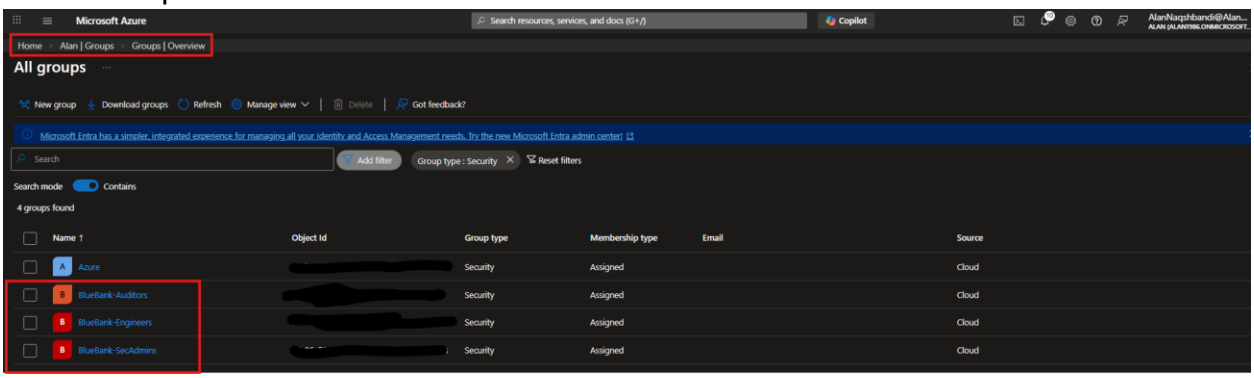
PHASE 0 – Resource Organization

Purpose: Establish governance, workload separation, and consistent tagging for cost tracking, ownership, and audit clarity.



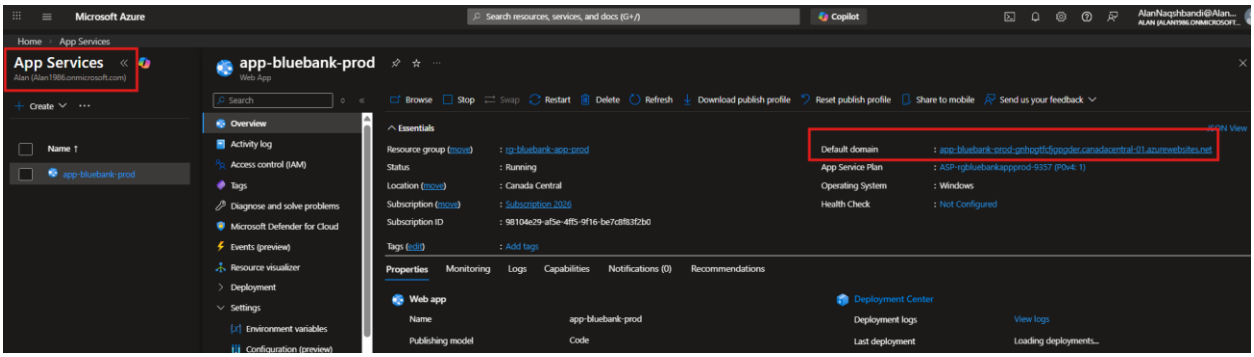
PHASE 1 – Identity & RBAC

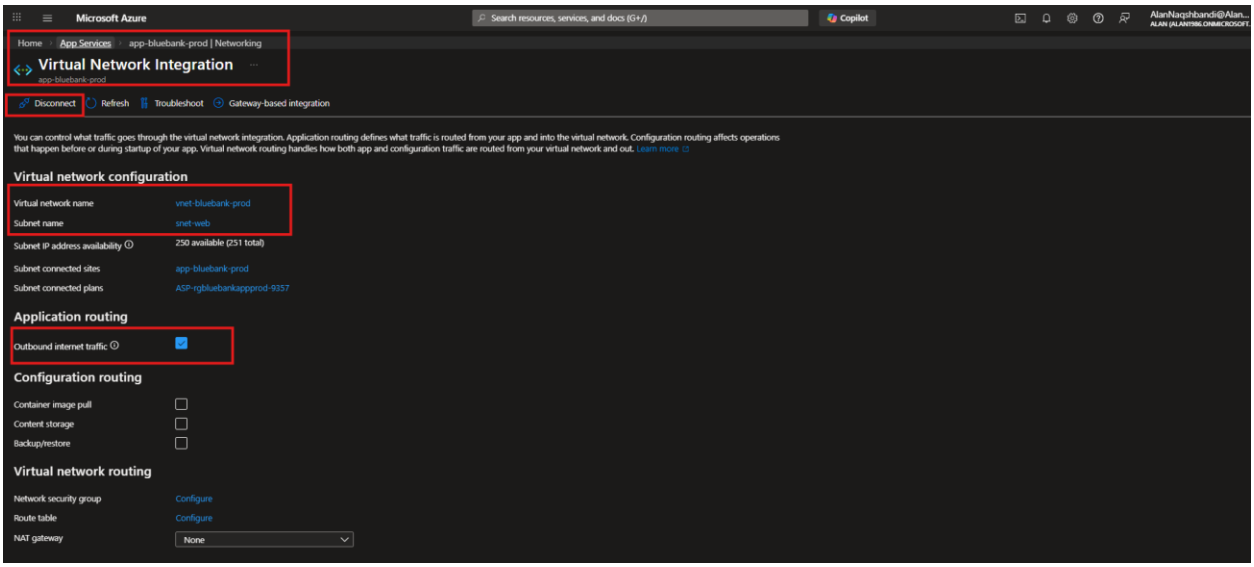
Purpose: Implement least-privilege access using Microsoft Entra ID security groups and RBAC at Resource Group scope, enforcing separation of duties and preventing direct user permissions.



PHASE 2 – Network Segmentation & Private Connectivity

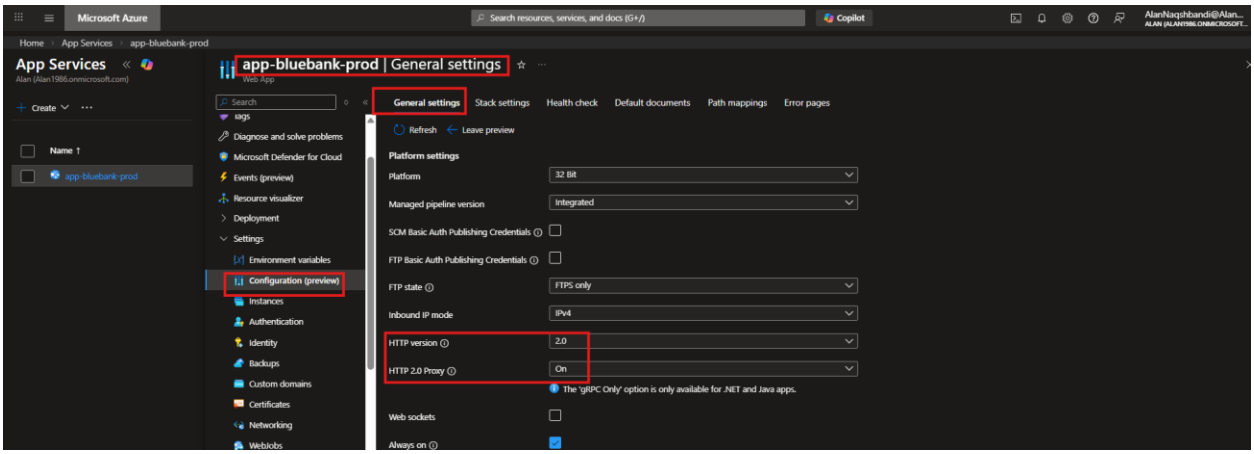
Purpose: Build the network foundation for a “public frontend / private backend” model using a segmented VNet, subnets, and Private DNS integration to support Private Endpoints securely.





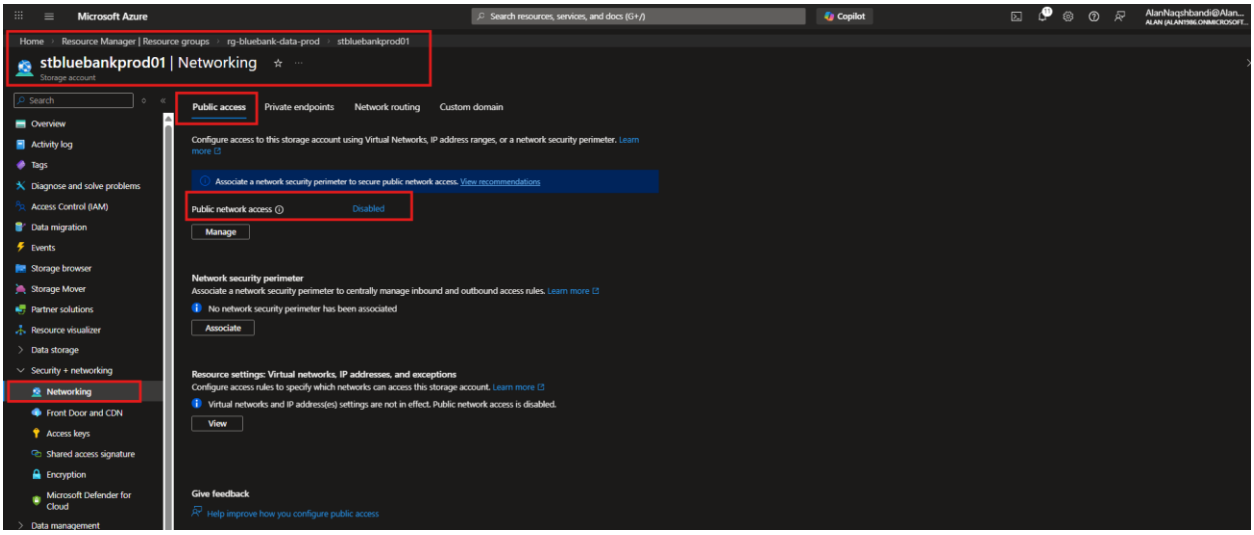
PHASE 3 – Application Layer (App Service)

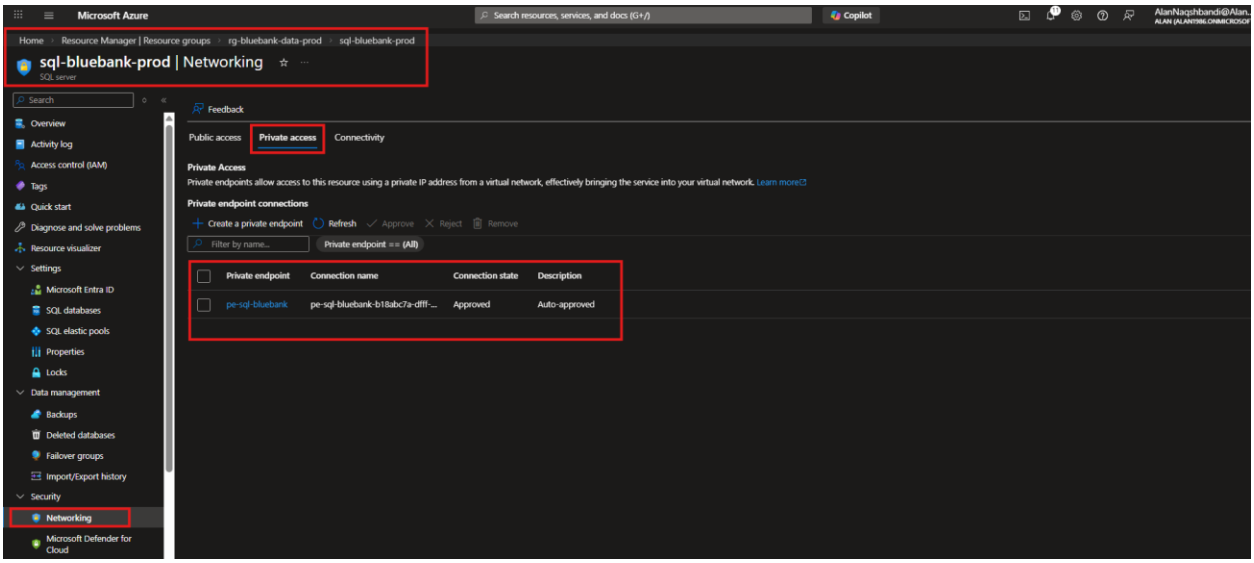
Purpose: Deploy the public-facing application tier with secure defaults (HTTPS-only), enable Managed Identity, and integrate the web app with the VNet to reach private backend services.



PHASE 4 – Secure SQL Deployment (Private + Protected)

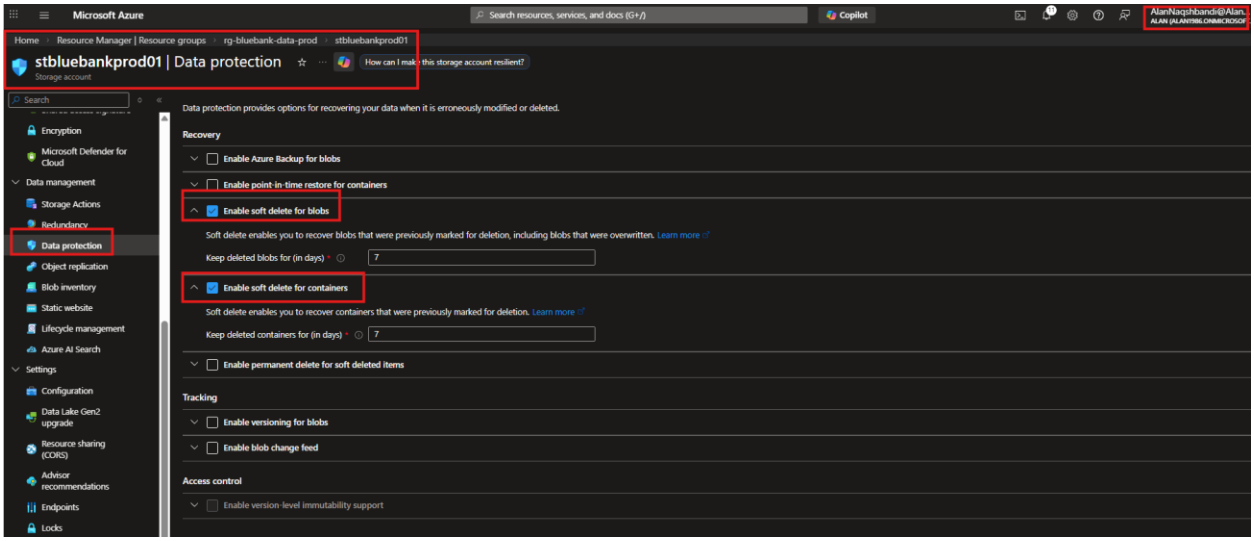
Purpose: Deploy the financial database with zero public exposure by disabling public access and enforcing private connectivity through Private Endpoint + Private DNS; enable Defender protections for database threat detection.





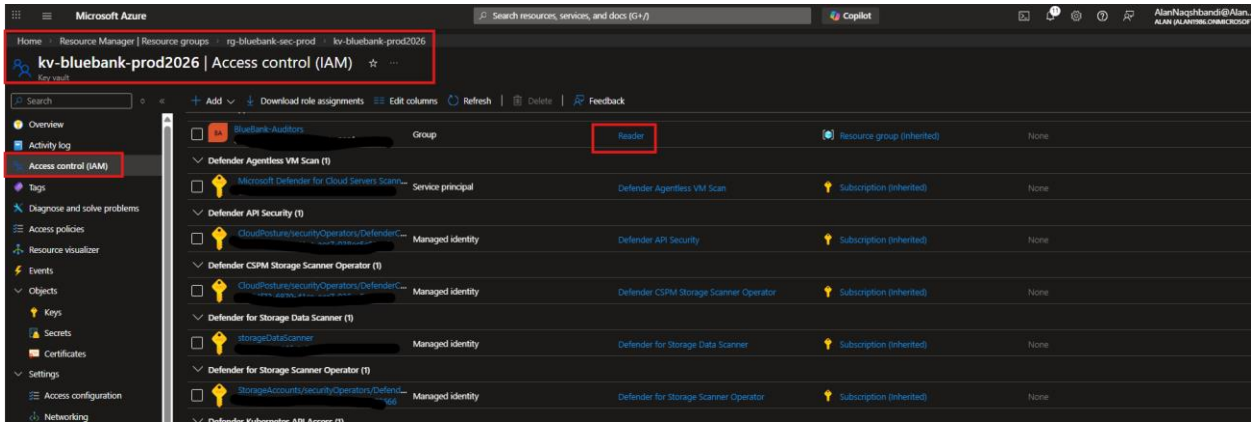
PHASE 5 – Secure Storage Deployment (Private + Data Protection)

Purpose: Deploy secure storage for financial documents with public access disabled, private connectivity via Private Endpoint, and resilience controls (soft delete + versioning); enable Defender protections for storage threat detection.



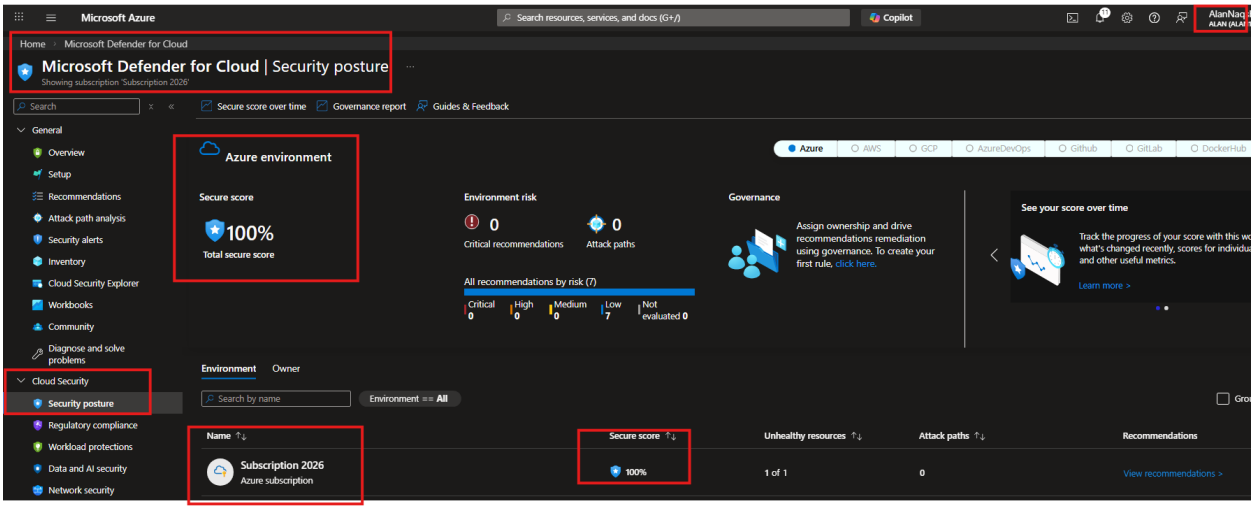
PHASE 6 – Secrets Management (Key Vault + Managed Identity)

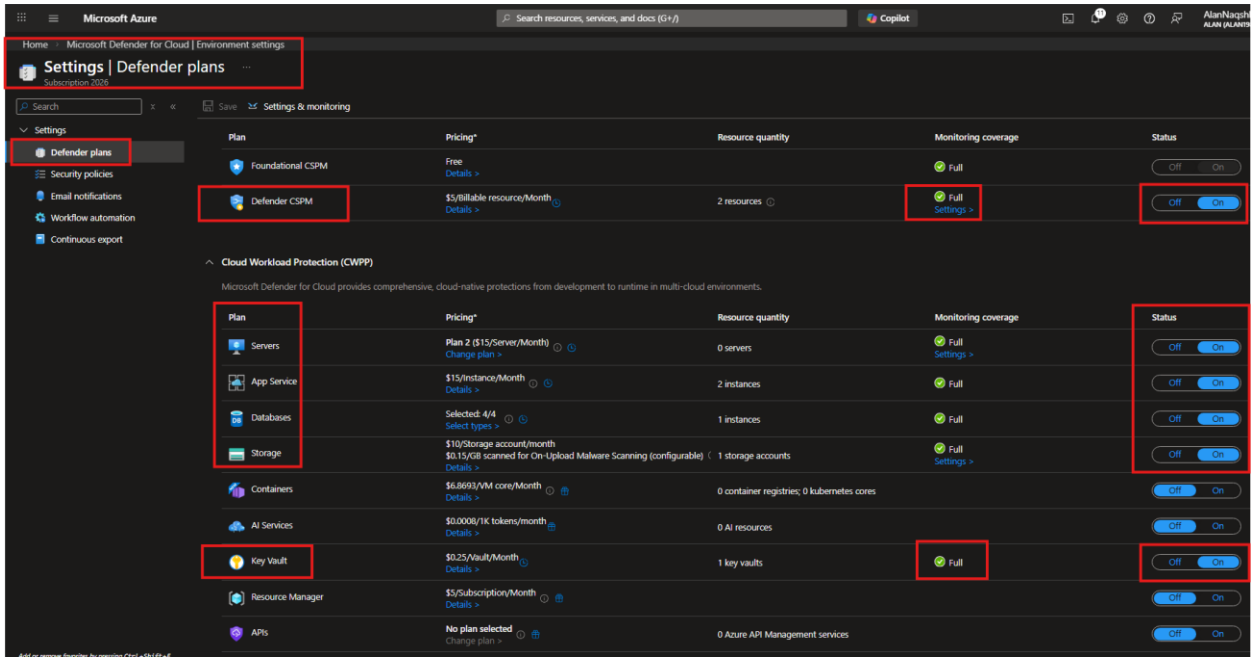
Purpose: Protect secrets using a private Key Vault (no public access), RBAC-based authorization, and Managed Identity for application access; store sensitive configuration such as database connection strings securely.



PHASE 7 – Monitoring, Alerting, and Security Posture

Purpose: Ensure operational visibility and readiness by centralizing logs in Log Analytics, enabling diagnostic settings across services, creating alert rules for availability/performance, and validating D efender for Cloud security posture.

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Microsoft Defender for Cloud provides comprehensive, cloud-native protections from development to runtime in multi-cloud environments.
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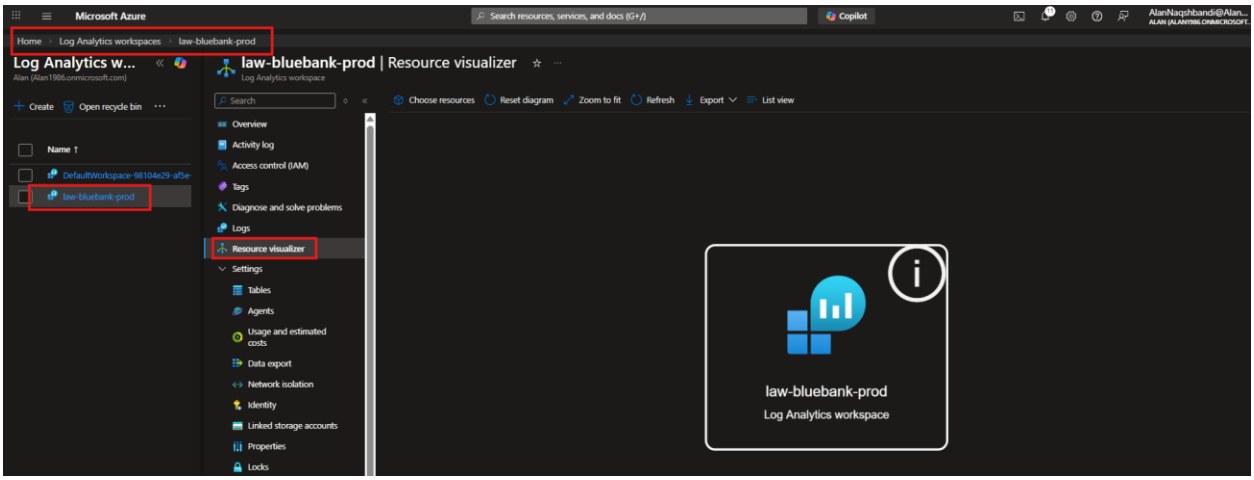
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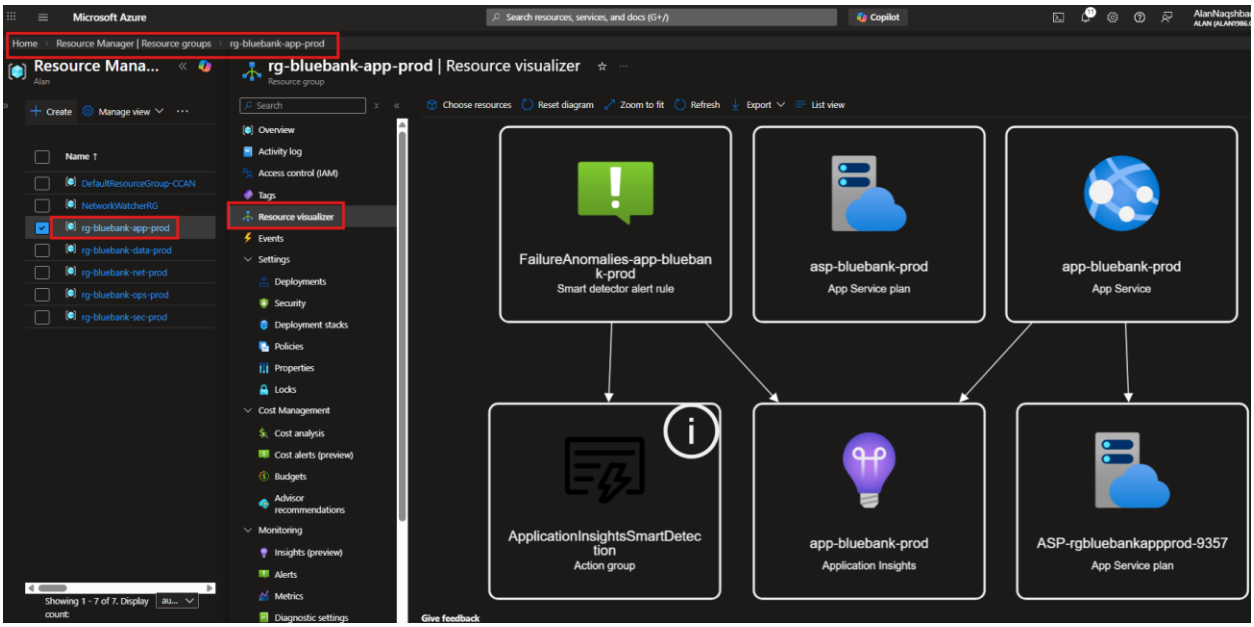
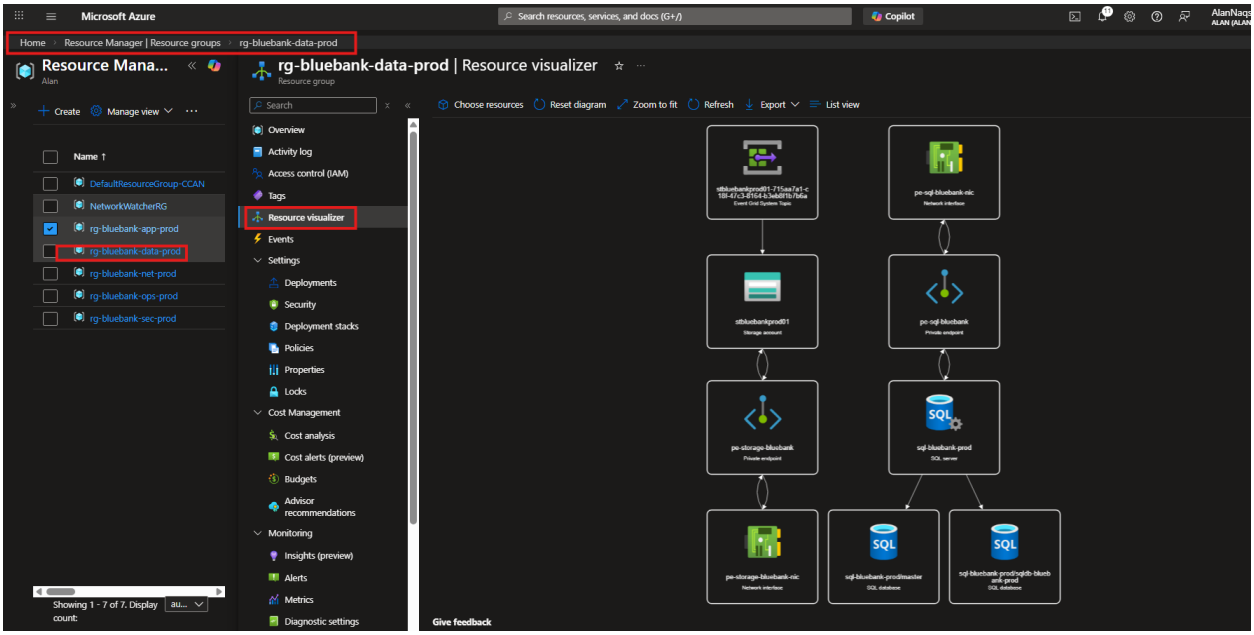
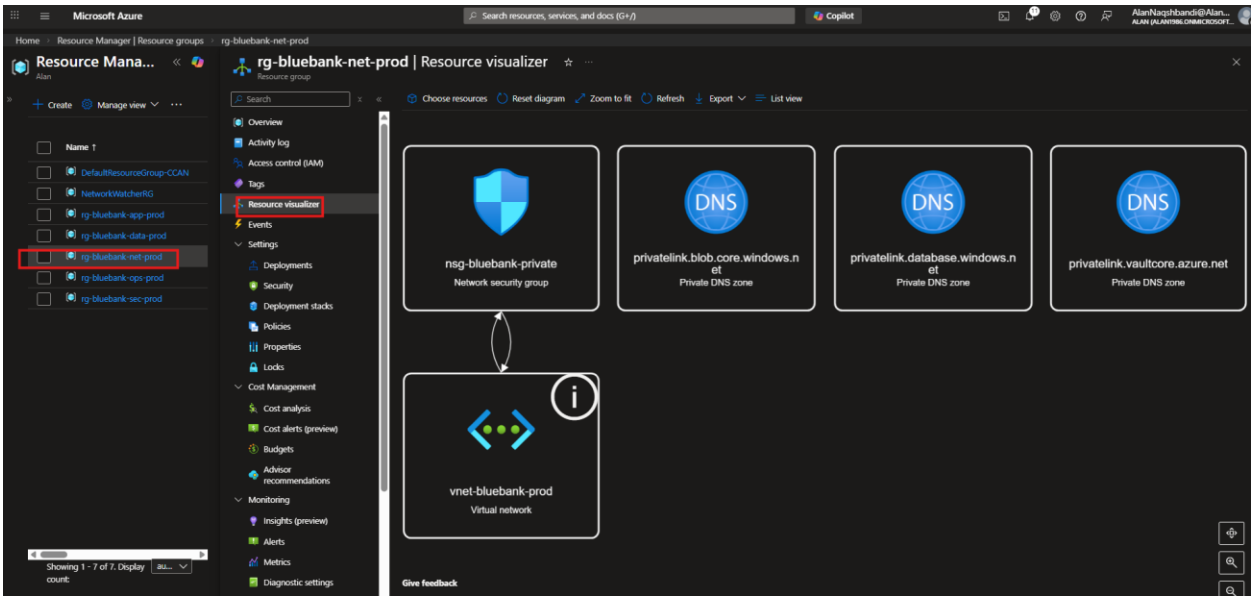
Choose resources
Reset diagram
Zoom to fit
Refresh
Export
List view
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Logs
Resource visualizer
Settings
Tables
Agents
Usage and estimated costs
Data export
Network isolation
Identity
Linked storage accounts
Properties
Locks
law-bluebank-prod
Log Analytics workspace

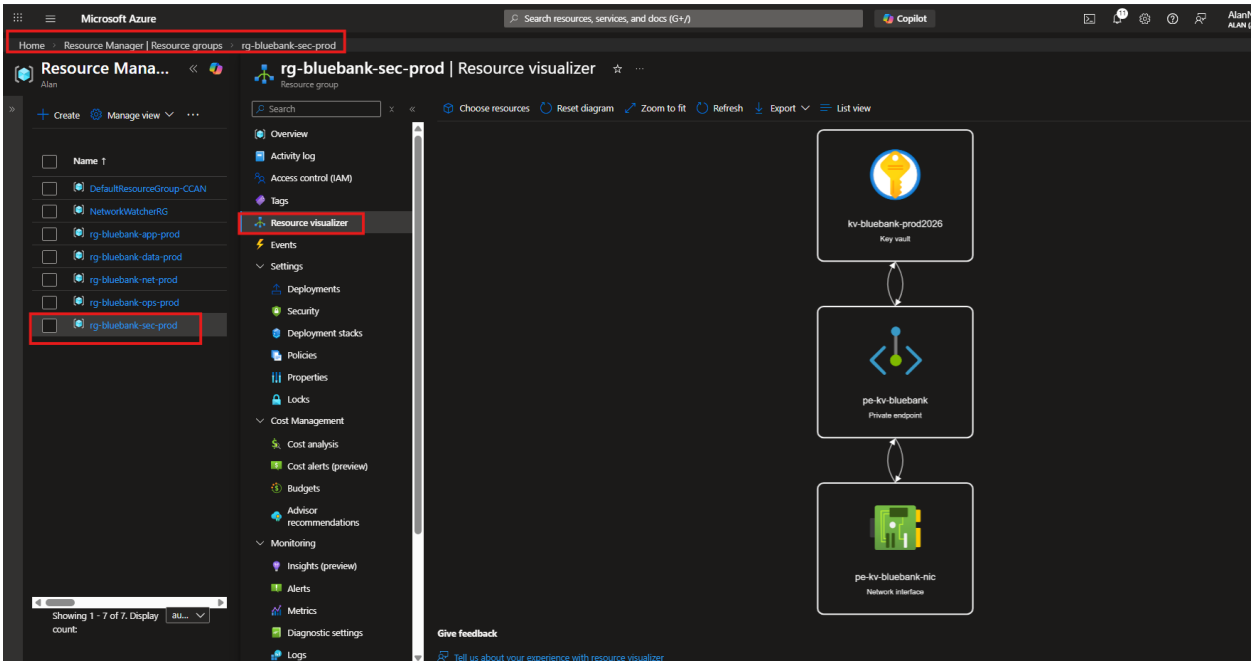
PHASE 8 – Production Validation

Purpose: Validate the end-to-end security posture by confirming no public exposure of backend services, verifying Private Endpoint approvals and DNS resolution, validating identity-based access paths, and confirming monitoring and alerts are operational.









🚀 Project Journey

This project demonstrates the end-to-end deployment of a secure, production-grade Azure environment designed for a financial services use case. It implements structured governance (resource groups, tagging, and RBAC), a segmented network foundation, private-only backend services through Private Endpoints and Private DNS, identity-based access via Managed Identity, and centralized monitoring with Log Analytics and Azure Monitor alerts. Security posture was strengthened using Microsoft Defender for Cloud protections for key services and validated through a final production review to confirm private exposure controls, access paths, and operational readiness aligned with real-world financial security standards.